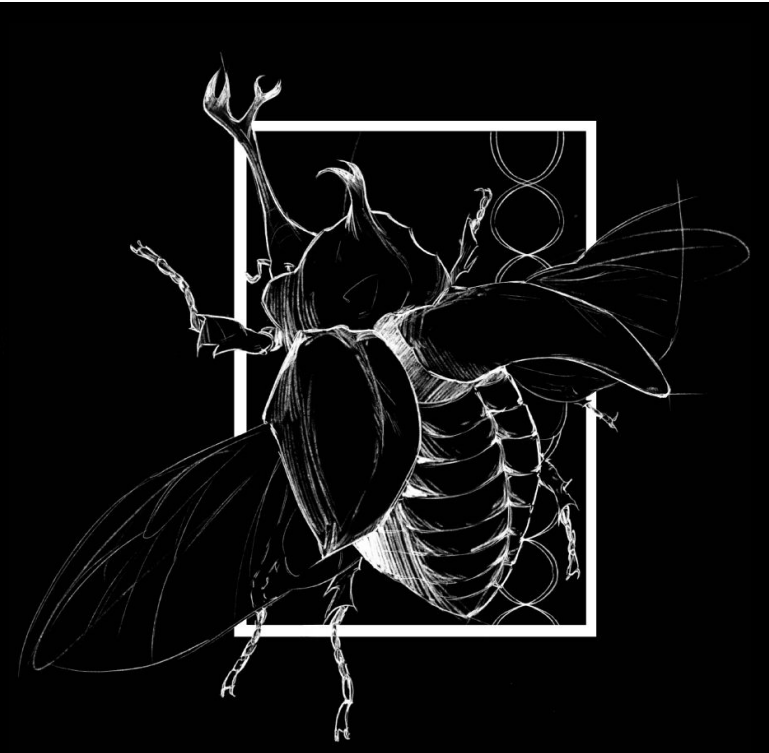




Effects of Developmental Strategy and Range Size on Chromosome Evolution in Salamanders

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Background

Chromosome number and structure influence genome organization, speciation, and diversification across the tree of life. Salamanders (order *Caudata*) show extensive variation in chromosome number, yet the drivers of this variation remain poorly understood. While prior work has focused on genome size and molecular mechanisms, less is known about how ecological and life-history traits shape chromosome evolution.

This study asks whether developmental strategy or geographic range size influence rates of chromosome evolution in salamanders.



Study Design & AI

This study uses a phylogenetic comparative framework to test whether developmental strategy and ecological context influence chromosome evolution in salamanders (*Caudata*). Species-level data on chromosome number, developmental strategy, and geographic range size were compiled and matched to a phylogenetic tree.

Developmental strategy was categorized as metamorphosing or paedomorphic, with facultative and obligate paedomorphs combined for analysis. And range size was used as a proxy for gene flow, genetic drift, and population isolation. Range size was classified into small (restricted/endemic), medium (regional), and large (widespread) categories and grouped into small versus medium/large for MCMC analysis.

Chromosome evolution was modeled across the phylogeny to test for differences in rates of fission, fusion, polyploidy, and trait transitions.

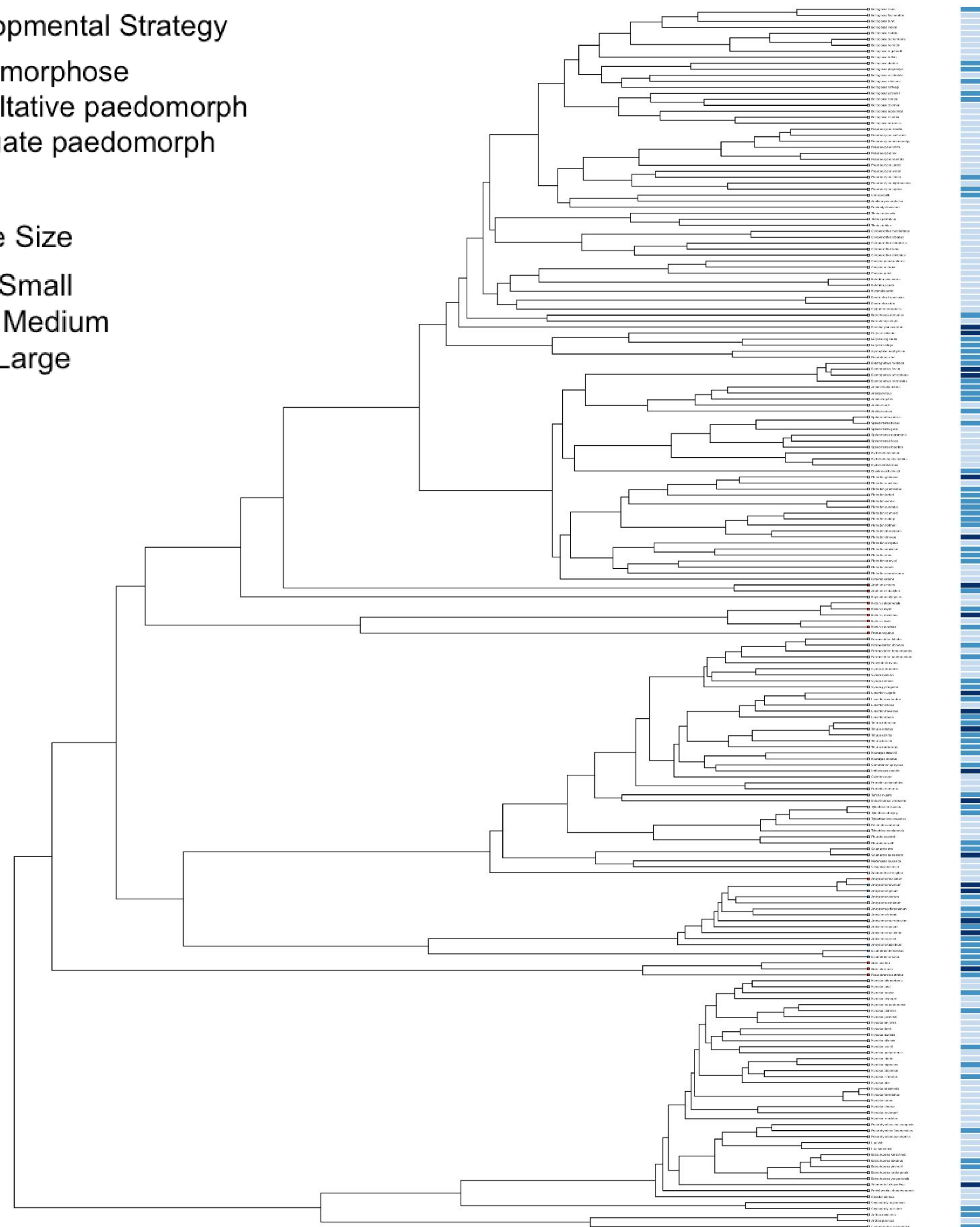
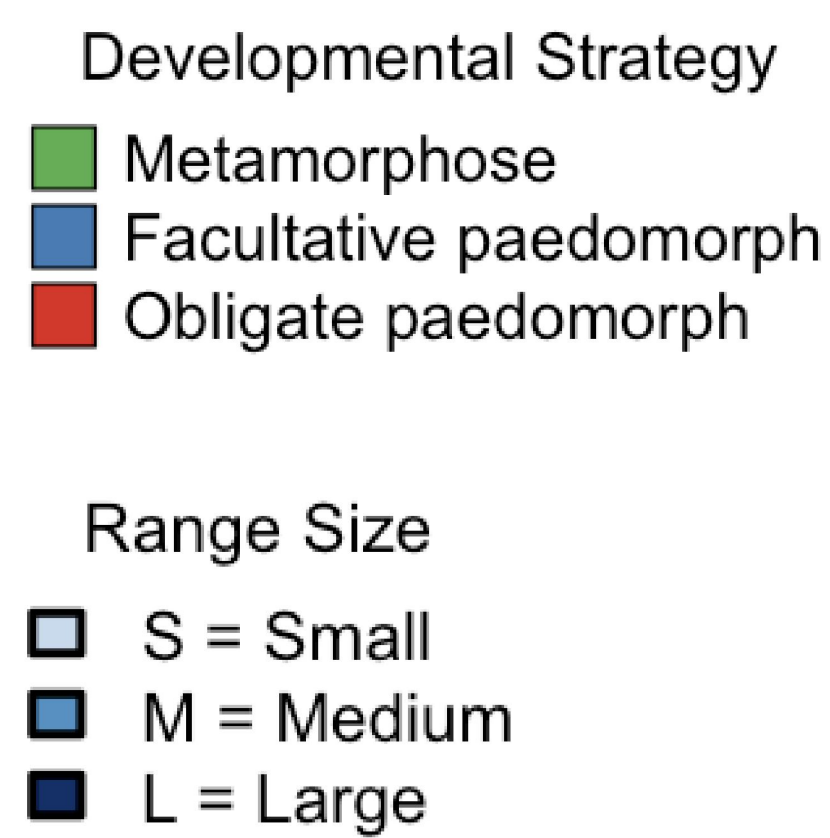
AI tools were used for data organization and collection, and coding support; all biological interpretations, trait classifications, and analytical decisions were reviewed and validated by the researcher.

Model Fitting

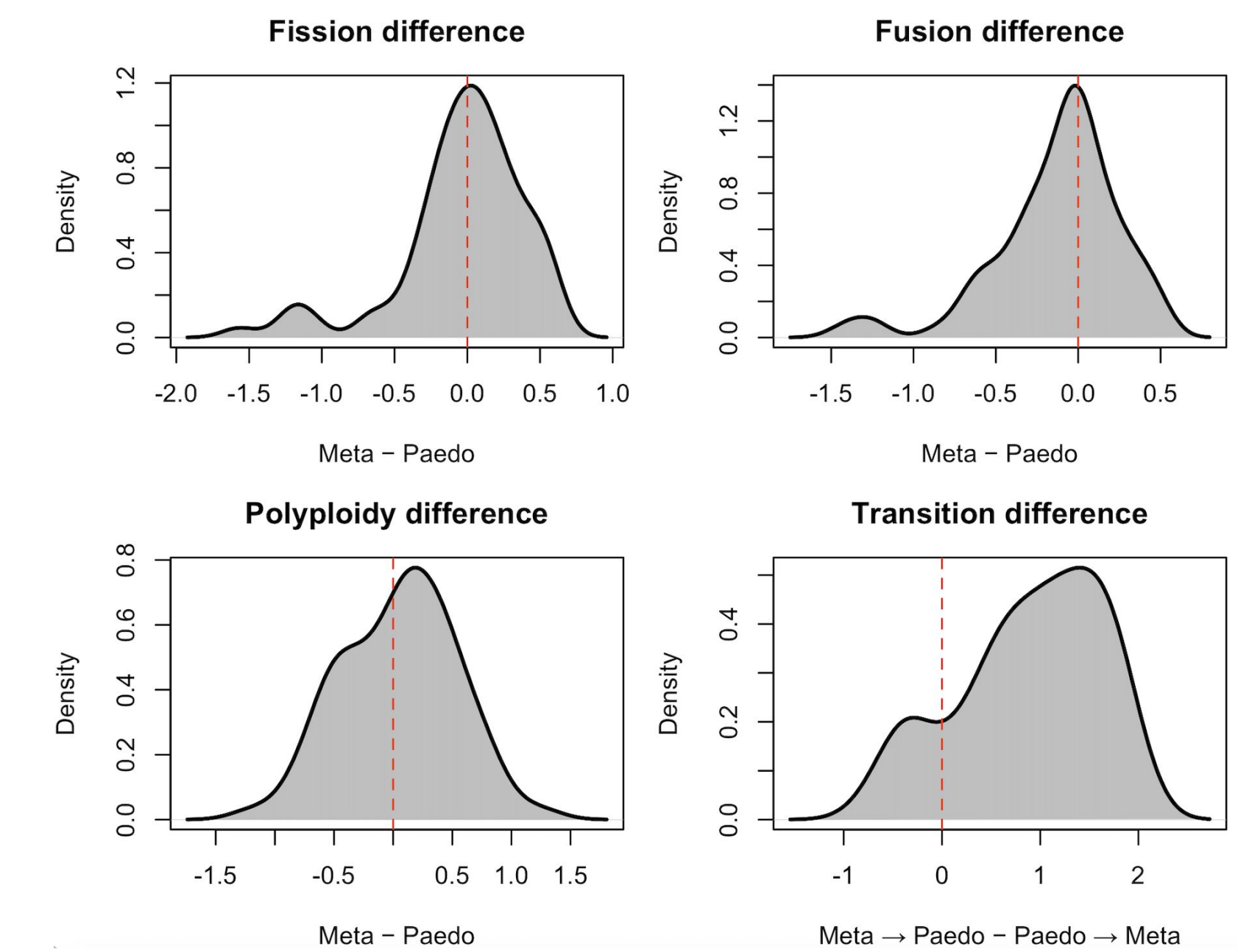
Models were fit using maximum likelihood and Markov Chain Monte Carlo (MCMC) methods to estimate posterior distributions of chromosome evolution parameters. A burn-in period was removed, and posterior samples were used to compare state-dependent rates of fission, fusion, polyploidy, and trait transitions.

Results

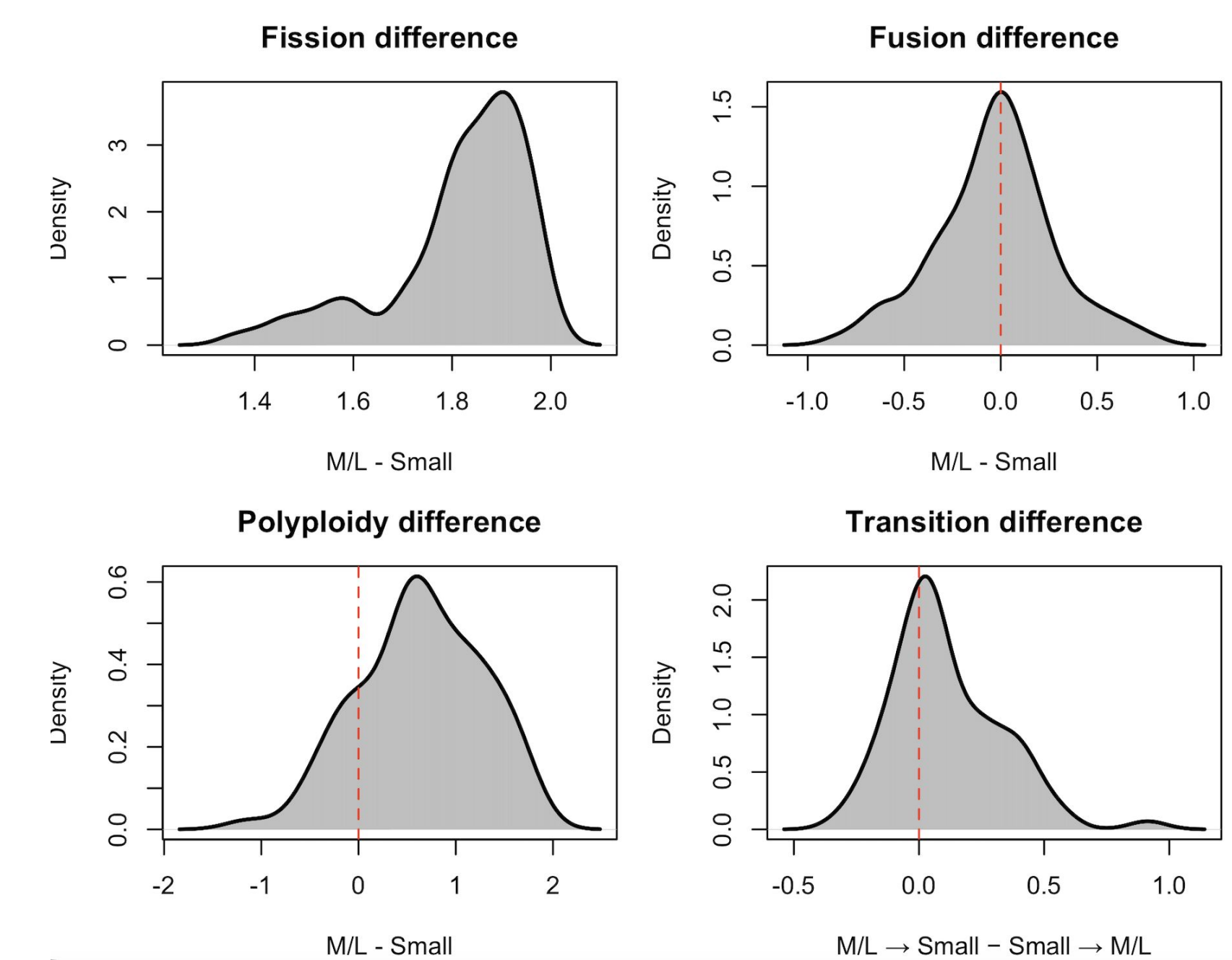
Phylogeny



Developmental strategy: Paedomorphosis vs Metamorphosis



Range size: Small vs Medium/Large



Results Analysis and Discussion

Results Analysis

Chromosome evolution rates showed substantial overlap between metamorphosing and paedomorphic species, indicating that developmental strategy has little to no effect on fission, fusion, or polyploidy rates. However, transition rates were asymmetric, with higher rates toward paedomorphosis than reversal.

In contrast, range size showed stronger effects. Fission rates were significantly higher in medium- and large-range species compared to small-range species, with minimal overlap in posterior distributions. Fusion rates showed no meaningful difference, while polyploidy rates were moderately lower in small-range species. These results indicate that range size influences specific components of chromosome evolution, particularly fission.

Discussion

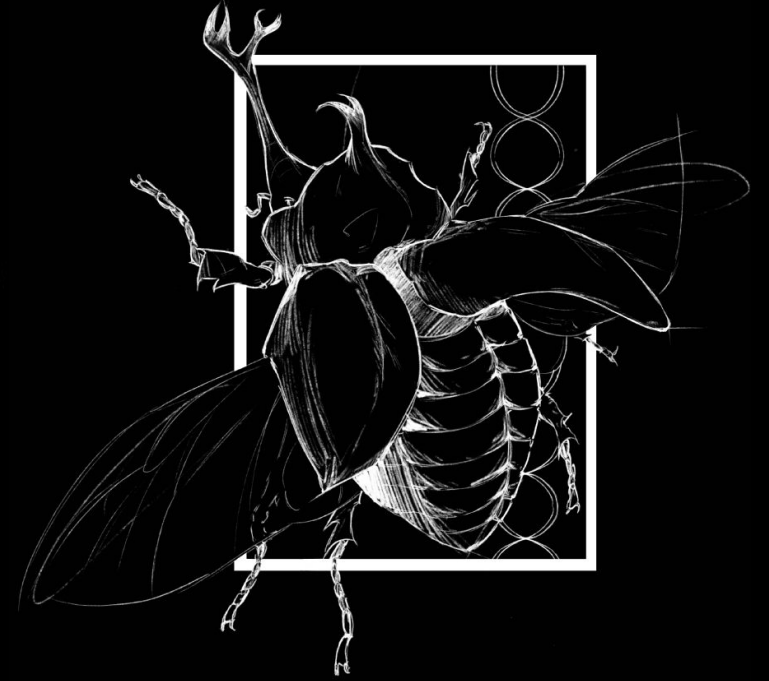
Chromosome evolution in salamanders appears to be more strongly influenced by ecological factors such as geographic range size than by developmental strategy. Even if chromosome numbers look similar, they evolve differently depending on natural history characteristic like range size..

Reduced fission rates in small-range species suggest that chromosome evolution may depend more on positive selection than on genetic drift alone. These findings highlight the importance of ecological context in shaping genome evolution.

Conclusions & Acknowledgements

Chromosome evolution in salamanders appears to be more strongly influenced by geographic range size than by developmental strategy. Even when chromosome numbers are similar, they evolve differently depending on the environment. These findings suggest that population size and genetic diversity in widespread species may play an important role in facilitating chromosome evolution.

Data sources included chromosome data from the Blackmon Lab, the IUCN Red List (for range information to determine range size category), and AmphibiaWeb (for developmental strategies. Computational resources and instruction were provided by Texas A&M University.



Background

Chromosome number and structure evolve in ways that influence genome organization, speciation, and diversification across the tree of life. Salamanders (order Caudata) exhibit extensive variation in chromosome number, yet the evolutionary drivers of this variation remain poorly understood. While previous work has focused on genome size and molecular mechanisms, less attention has been given to how ecological, geographical and life history traits shape chromosome evolution.

So the main question is: Do life-history traits (developmental strategy) or ecological factors (range size) influence the rates of chromosome evolution in salamanders?



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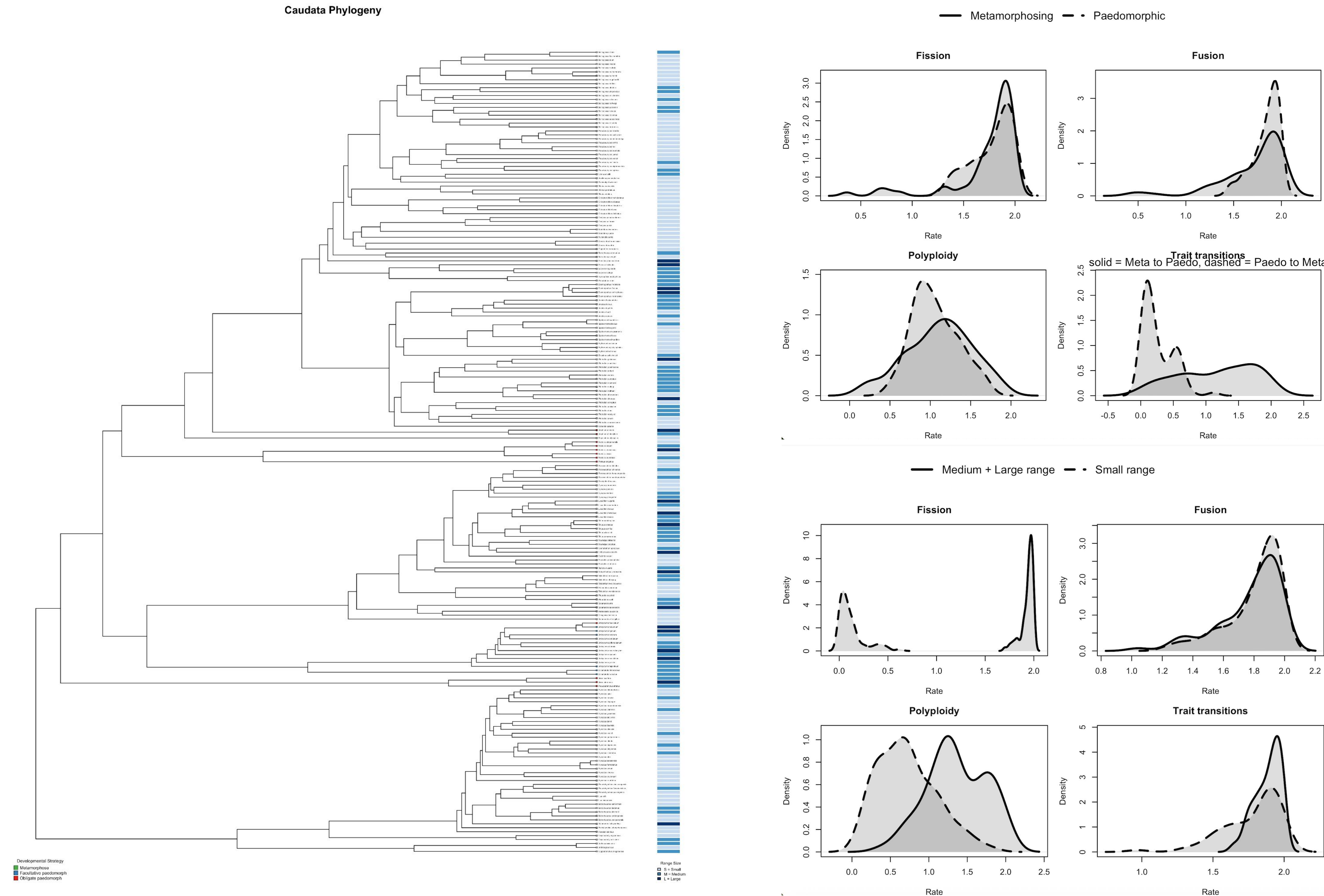
Developmental strategy was categorized as metamorphosing or paedomorphic, and range size was used as a proxy for gene flow and population isolation. Range size was classified into small, medium, and large categories based on species distribution data and grouped into small versus medium/large categories for MCMC analysis. Chromosome evolution was modeled across the phylogeny to test whether these traits are associated with differences in evolutionary rates.

Artificial intelligence tools were used to assist with data organization, compiling data from databases and verified websites, writing R scripts and workflow development. All biological interpretations, trait classifications, and analytical decisions were reviewed and validated

Model Fitting

Models were fit using maximum likelihood and Markov Chain Monte Carlo (MCMC) methods to estimate posterior distributions of chromosome evolution parameters. A burn-in period was removed, and posterior samples were used to compare state-dependent rates of fission, fusion, polyploidy, and trait transitions.

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In contrast, range size showed stronger associations with chromosome evolution dynamics. Fission rates were markedly higher in medium- and large-range species compared to small-range species, with minimal overlap in posterior distributions. Fusion rates showed little difference across groups, while polyploidy rates were moderately lower in small-range species. These results indicate that range size is associated with specific components of chromosome evolution.

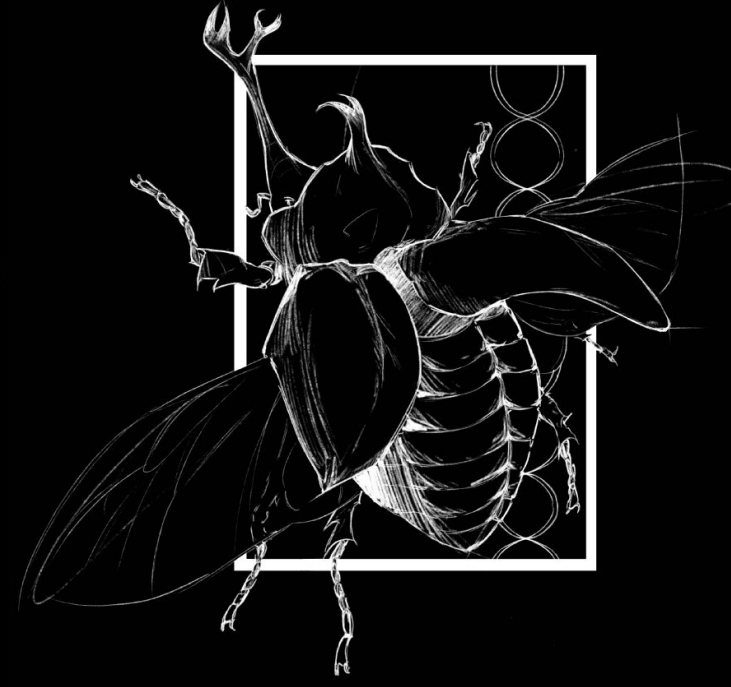
Discussion

Originally, Different it was expected that ecological lifestyles (developmental strategy) could influence how chromosomes evolve by influencing genetic drift and flow. Developmental strategy showed limited influence on chromosome evolution, suggesting that differences between metamorphosing and paedomorphic life histories do not strongly impact chromosomal processes. However, asymmetric transition rates indicate that paedomorphosis may evolve more readily than it is lost.

In contrast, range size was also expected to affect chromosome evolutions and the results showed that it is in fact more strongly associated with chromosome evolution dynamics. Small-range species exhibited substantially reduced fission rates, suggesting that chromosomal restructuring may occur less frequently in geographically restricted taxa. This pattern does not support the expectation that increased genetic drift accelerates chromosome evolution. Instead, it suggests that broader geographic distributions may facilitate higher rates of chromosomal change.

Conclusions & Acknowledgements

Chromosome evolution in salamanders appears to be more strongly associated with ecological factors such as geographic range size than with developmental strategy. While chromosome number itself may be conserved, underlying evolutionary processes differ across ecological contexts. These findings suggest that population and genomic dynamics in widespread species may play a key role in facilitating chromosome evolution. Citations.....



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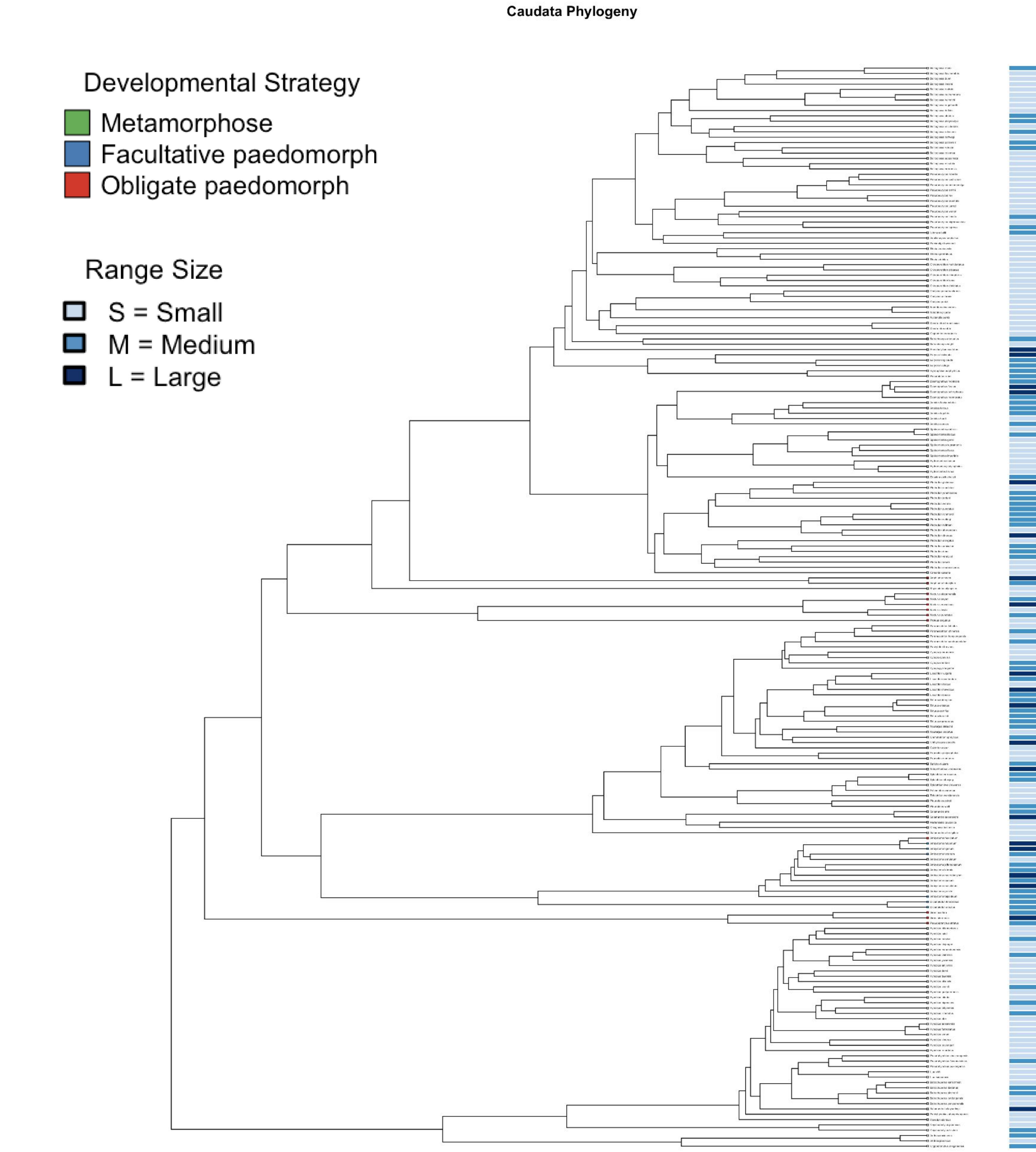
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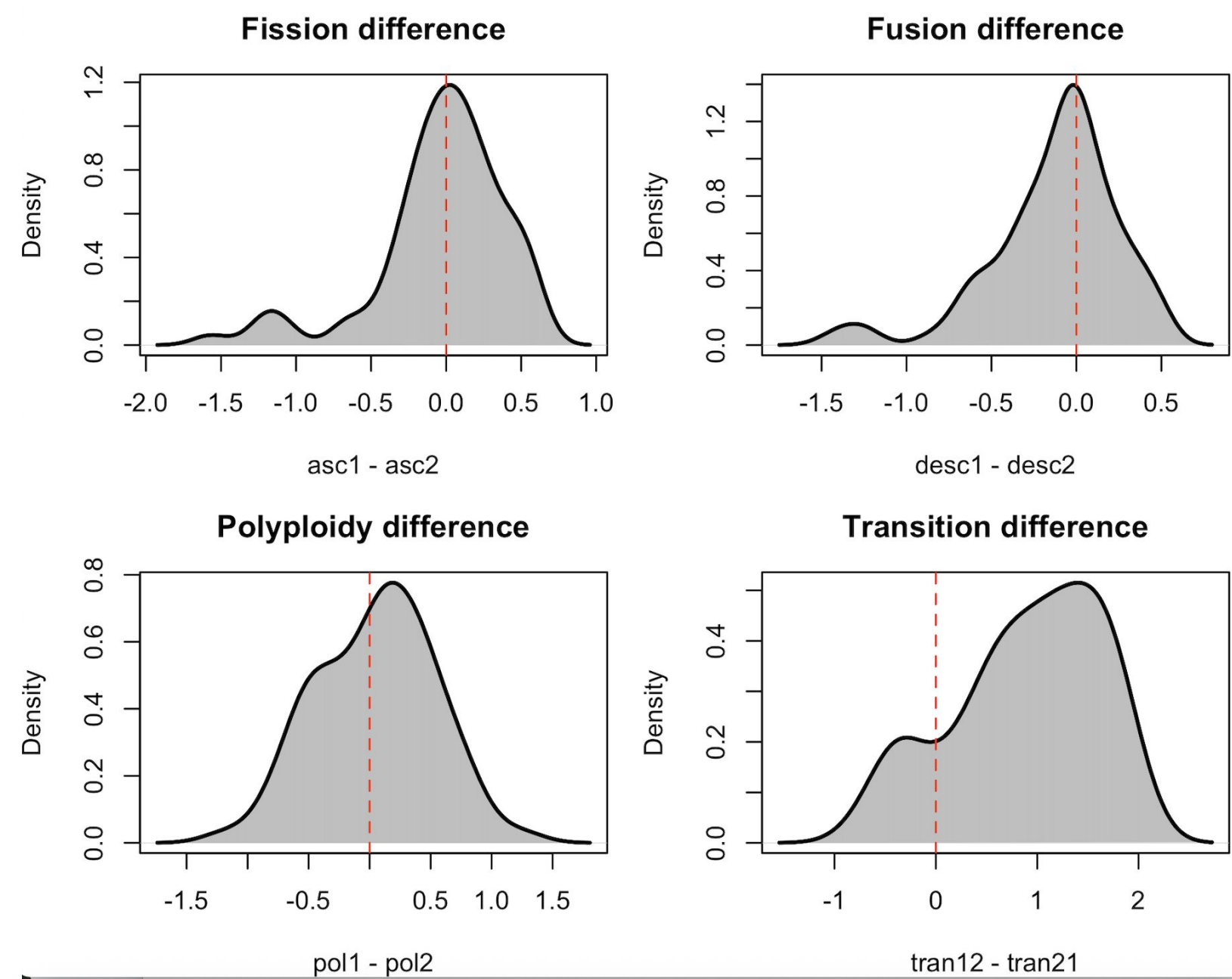
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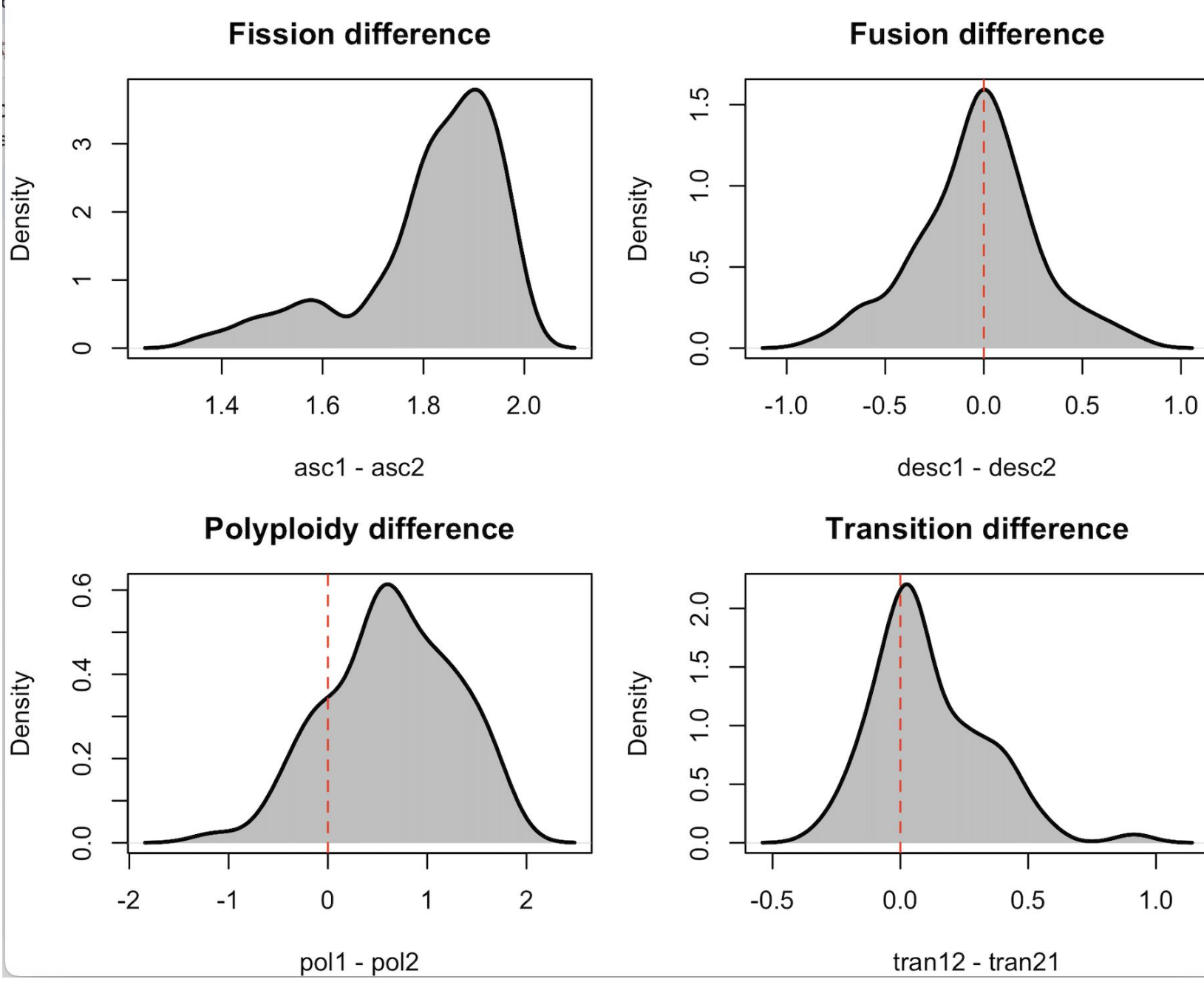
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Range size: Small vs Medium/Large



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